

Liam Dubai

Ph.D. Candidate in Astronomy
The Ohio State University

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Education

The Ohio State University	Columbus, OH
Ph.D. Candidate in Astronomy	2024 – Present
M.S. in Astronomy	August 2024
Thesis Advisor: Prof. Jennifer A. Johnson	
Whitman College	Walla Walla, WA
B.A. in Physics-Astronomy, Music (Performance), <i>summa cum laude</i>	May 2021
Honors in Physics-Astronomy, Thesis Advisor: Prof. Andrea K. Dobson	
Honors in Music (Performance), Thesis Advisor: Dr. Sally Singer Tuttle	

Honors & Awards

OSU Presidential Fellowship	2026–2027
Support for final-year students in recognition of outstanding scholarly accomplishments.	
NSF Graduate Research Fellowship Proposal , Honorable Mention	2022
The Ohio State University Graduate Fellowship	2021 – 2022
Financial support for first-year graduate students.	
Whitman College Scholarships	
Campbell Music Scholarship	2017 – 2021
Walter Brattain Scholarship	2017
William O. Douglas Valedictorian Scholarship	2017

Research

Interests: Galactic chemical evolution, the Milky Way, origin of the elements, Type Ia supernovae, large stellar spectroscopic surveys

Origin and Enrichment of Neutron-Capture Elements	2025 – Present
The Ohio State University	
Advisors: Prof. Jennifer A. Johnson, Prof. Keith Hawkins, Dr. James W. Johnson	
Constraining the Milky Way’s Accretion History	2024 – 2025
The Ohio State University	
Advisors: Prof. Jennifer A. Johnson, Dr. James W. Johnson	

- Timescale of Iron Production in Galaxy Evolution Models** 2021 – 2024
 The Ohio State University
 Advisors: Prof. Jennifer A. Johnson, Dr. James W. Johnson
- Rare Supernovae in Archival Space Telescope Data** 2020 – 2021
 University of Hawai‘i Institute for Astronomy, NSF REU program
 Advisors: Prof. Benjamin J. Shappee, Dr. Michael A. Tucker
- Instrumentation Noise of Space-Based Gravitational Wave Detectors** 2019
 The University of Alabama in Huntsville /
 NASA Marshall Space Flight Center, NSF REU program
 Advisor: Dr. Tyson Littenberg

Publications

NASA ADS Library of Publications

- Dubay, L. O.** et al. (16 additional authors) 2026, “A Broken Clock Is Right Twice a Day: [Ce/Mg] Is Not a Universal Chemical Clock”, *arXiv e-prints*, [arXiv:2608.26149](https://arxiv.org/abs/2608.26149)
- SDSS Collaboration et al. (inc. **Dubay, L.**) 2026, “The Twentieth Data Release of the Sloan Digital Sky Survey: First All-Sky BOSS Spectra, eROSITA-SDSS-V Mapper Coordinated Observations, and a Preview of the Local Volume Mapper”, *arXiv e-prints*, [arXiv:2607.26149](https://arxiv.org/abs/2607.26149)
- Roberts, J. D., Pinsonneault, M. H., Johnson, J. A., **Dubay, L. O.**, & Johnson, J. W. 2026, “[C/N] Ages for Red Giants and their Implications for Galactic Archaeology”, *The Astrophysical Journal* 1002, 191, doi:[10.3847/1538-4357/ae5bae](https://doi.org/10.3847/1538-4357/ae5bae)
- Dubay, L. O.**, Johnson, J. A., Johnson, J. W., Roberts, J. D. 2026, “Challenges to the Two-Infall Scenario by Large Stellar Age Catalogs”, *The Astrophysical Journal* 1000, 244, doi:[10.3847/1538-4357/ae4c53](https://doi.org/10.3847/1538-4357/ae4c53)
- Johnson, J. W. et al. (inc. **Dubay, L. O.**) 2025, “The Milky Way Radial Metallicity Gradient as an Equilibrium Phenomenon: Why Old Stars Are Metal Rich”, *The Astrophysical Journal* 988, 8, doi:[10.3847/1538-4357/addbe5](https://doi.org/10.3847/1538-4357/addbe5)
- Dubay, L. O.**, Johnson, J. A., & Johnson, J. W. 2024, “Galactic Chemical Evolution Models Favor an Extended Type Ia Supernova Delay-Time Distribution”, *The Astrophysical Journal* 973, 55, doi:[10.3847/1538-4357/ad61df](https://doi.org/10.3847/1538-4357/ad61df)
- Dubay, L. O.**, Tucker, M. A., Do, A., Shappee, B. J., & Anand, G. S. 2022, “Late-Onset Circumstellar Medium Interactions are Rare: An Unbiased *GALEX* View of Type Ia Supernovae”, *The Astrophysical Journal* 926, 98, doi:[10.3847/1538-4357/ac3bb4](https://doi.org/10.3847/1538-4357/ac3bb4)

Talks & Posters

Contributed Talk	Sloan Digital Sky Survey (SDSS-V) Collab. Meeting (Leuven, Belgium)	2026
Invited Seminar	Carnegie Observatories (Pasadena, CA)	2026
Poster	SDSS-V Collaboration Meeting (Heidelberg, Germany)	2025
Contributed Talk	SDSS-V Collaboration Meeting (Las Cruces, NM)	2024
Contributed Talk	Hayes Advanced Research Forum , Ohio State University	2024
Poster	Surveying the Milky Way: The Universe in Our Own Backyard , California Institute of Technology (Pasadena, CA)	2023
Contributed Talk	241st American Astronomical Society (AAS) Meeting (Seattle, WA)	2023
Poster	237th AAS Meeting (Virtual)	2021
Poster	235th AAS Meeting (Honolulu, HI)	2020

Teaching

Polaris Mentorship Course Instructor, The Ohio State University 2025 – 2026
 Physics 2050, *Instructor of record*

Year-long course that pairs first- and second-year physics undergraduates with graduate student mentors; topics include study skills, science identity, and impostor syndrome; introduces students to physics research with a guided project and poster presentation.

Teaching Coursework

Astronomy 7193: College Teaching in Astronomy Summer 2025

Graduate Teaching Assistant, The Ohio State University 2022 – 2023

Astronomy 3350: Methods of Astronomical Observation & Data Analysis Fall 2023

Astronomy 1221: Astronomy Data Analysis (2 semesters) Spring – Fall 2023

Astronomy 1101: Planets to Cosmos Lab (2 sections) Fall 2022

Astronomy Teaching Assistant & Tutor, Whitman College 2019 – 2020

Astronomy 177: Sky and Planets Spring 2020

Astronomy 179: Galaxies and Cosmology Fall 2019

Astronomy 178: Sun and Stars Spring 2019

Service & Mentoring

Polaris Leadership Committee, The Ohio State University 2023 – Present

Graduate student-led organization dedicated to fostering community and improving retention of non-traditional and underrepresented students in physics and astronomy.

Website: <https://u.osu.edu/polaris>

“Galaxy Hour” Meeting Co-Organizer, Ohio State Astronomy Dept. 2023 – Present
Organize weekly topical research meeting and visiting speaker seminars.

Python Coding Workshops Lead Instructor 2024 – 2025
Organized lectures and developed engaging challenge problems on Python fundamentals for undergraduate summer research students.

Near-Peer Mentor, Polaris Mentorship Course 2022 – 2025
Mentored four first- and second-year undergraduates from underrepresented backgrounds during a year-long course; lead guided research projects culminating in poster presentations.

Outreach

Planetarium Presenter, Arne Slettebak Planetarium 2021 – 2025
The Ohio State University, Columbus, OH
Lead presenter for over 100 planetarium shows.
Audiences include: K-12, boy & girl scouts, university groups, and the general public.

Library Talks, Columbus Metropolitan Libraries Summer 2023
Astronomy talks for elementary school audiences.